DATABASES

Query Planning Visual Guide

EXPLAIN ANALYZE, index selection, join order -- think like the planner.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Reading EXPLAIN ANALYZE

A tree of nodes: read bottom-up, inside-out.

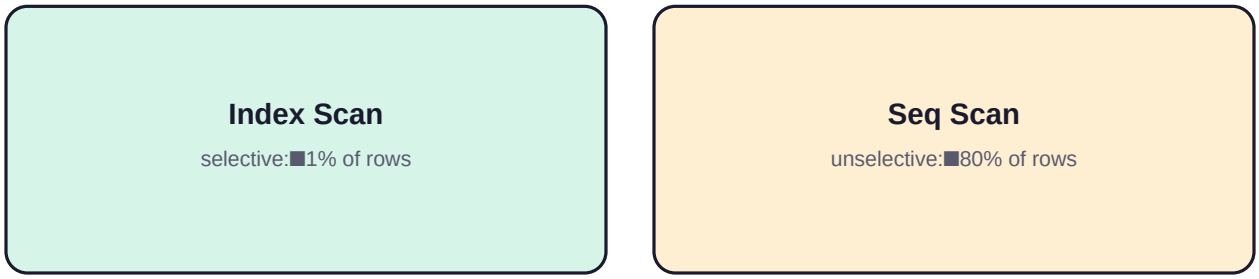
```
Hash Join  (cost=...) (actual rows=8421 loops=1)
  ->  Seq Scan on orders      (actual rows=50000)
  ->  Hash  (actual rows=1200)
        ->  Index Scan on customers (actual rows=1200)
```

- Indentation = execution order: children run before parents.
- actual rows vs estimated rows: a 10x gap means stale statistics.
- loops > 1: the node ran repeatedly -- multiply cost by loops.
- ANALYZE runs the query (use EXPLAIN without ANALYZE to avoid executing).

STEP 2/6

Index Selection

The planner is a cost accountant, not a rule follower.



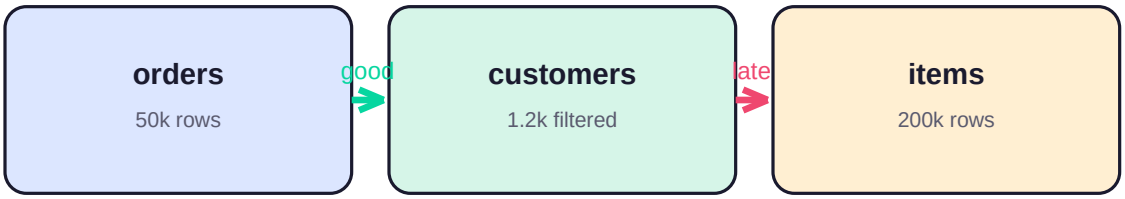
- An index is skipped when the planner expects to read most of the table anyway.
- Random page cost vs seq page cost: indexes win on selectivity, lose on bulk.
- Functions on columns (WHERE lower(email)=...) defeat indexes -- index the expression instead.

The planner can be wrong: if it misestimates selectivity, it picks the wrong scan. Fix the stats, not the query.

STEP 3/6

Join Ordering

Join the most selective tables first.



- Small intermediate results keep hash tables in memory -- join order is memory order.
- The planner reorders joins freely (join_collapse_limit); FROM order is a hint at best.
- Cross joins (missing ON clause) explode cardinality -- the classic accidental cartesian product.

STEP 4/6

Statistics

The planner only knows what ANALYZE told it.

- Stale stats after bulk loads: run ANALYZE manually, don't wait for autovacuum.
- Skewed columns (status='active' 99%) need higher statistics targets.
- Correlated columns (city, zip) fool single-column stats -- extended statistics help.

STEP 5/6

Fixing Slow Queries

A checklist, in order.

1. EXPLAIN ANALYZE

find the node eating time -- don't guess

2. Missing index?

high-cost seq scan on a selective filter

3. Wrong data type

joining text to int defeats indexes and hashes

4. N+1 queries

one query per row in a loop -- batch or JOIN it

STEP ✓I✓

Key Takeaways

Query Planning -- Remember

- Read EXPLAIN bottom-up: children execute first; compare actual vs estimated rows.
- The planner picks scans by cost: selective filters get indexes, bulk reads get seq scans.
- Join selective tables first -- small intermediates keep hash joins in memory.
- Statistics drive every decision: ANALYZE after bulk loads, raise targets on skewed columns.
- Fix slow queries in order: EXPLAIN, missing index, type mismatches, N+1 patterns.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev